

MTH344 Sample Quiz 1 (100 pts + EC 20 pts)

Please print your name clearly. A penalty may be applied if your name is difficult to read.

Last Name: _____

First Name: _____

Instructions:

- There are six problems. The points for each problem are indicated next to the question.
- To earn full credit, you must show all necessary work. If it is unclear which assumptions or theorems may be used without proof, write out as much of your reasoning as possible.
- This test is closed-notes, and no computing devices (including calculators) are allowed.

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P1 (20 pts)

Let $X \sim \text{Gamma}(\alpha, \lambda)$.

a. (10 pts) Show that, if $\alpha \leq 1$, the corresponding PDF is a convex function.

b. (10 pts) Show that, if $\alpha > 1$, the PDF is maximized at $x = (\alpha - 1)/\lambda$.

P2 (20 pts)

For a given $\lambda > 0$, let $(X, Y)^T$ have the joint PDF given by:

$$f(x, y) = \begin{cases} \lambda^2 e^{-\lambda y}, & 0 \leq x \leq y, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Find the conditional PDF of X given Y , $f_{X|Y}(x | y)$.

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P3 (20 pts)

Let $X_n \sim \text{Bin}(n, p_n)$ be a sequence of binomial random variables. Show that if $\lim_{n \rightarrow \infty} p_n = 0$ and $\lim_{n \rightarrow \infty} np_n = \lambda > 0$, then for any fixed $x \in \{0, 1, \dots\}$,

$$\lim_{n \rightarrow \infty} p_{X_n}(x) = \frac{\lambda^x}{x!} e^{-\lambda}. \quad (2)$$

P4 (20 pts)

Let $\Theta \sim \mathcal{U}(0, 1)$ and $X \mid \Theta = \theta \sim \text{Bin}(n, \theta)$. Show that the conditional distribution of Θ given X , the *posterior distribution*, is $\Theta \mid X = x \sim \text{Beta}(x + 1, n - x + 1)$.

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P5 (20 pts)

Let

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix} \right), \quad (3)$$

where $(\mu_X, \mu_Y)^T \in \mathbb{R}^2$, $\sigma_X > 0$, $\sigma_Y > 0$, and $-1 < \rho < 1$. Show that $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and

$$Y \mid X = x \sim \mathcal{N} \left(\mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X), \sigma_Y^2 (1 - \rho^2) \right). \quad (4)$$

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P6 (EC 20 pts)

Let $\boldsymbol{\mu} \in \mathbb{R}^d$. Let $\Sigma \in \mathbb{R}^{d \times d}$ be a symmetric and positive definite matrix with spectral decomposition $\Sigma = U\Lambda U^T$, where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ and $U = [\mathbf{e}_1 \mid \dots \mid \mathbf{e}_d]$. Show that if we represent a vector $\mathbf{x}$ as $\mathbf{x} = \boldsymbol{\mu} + \sum_{i=1}^d \sqrt{\lambda_i} z_i \mathbf{e}_i$, then the following factorization holds:

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \Sigma) = \prod_{i=1}^d \mathcal{N}(z_i; 0, 1). \quad (5)$$